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Two-piece spine implant installation instrument for use with an endoscope and method of use

Abstract

A two-piece medical instrument for installing an expandable interbody (intervertebral) spine implant via an endoscope, has a handle assembly and a shaft assembly configured for extension through the endoscope, the shaft assembly holding an expandable interbody spine implant when the shaft assembly is fully received by a head of the handle assembly, and releasing the spine implant upon retraction of the shaft assembly from the head of the handle assembly. The shaft assembly has a rod with a longitudinal bore and a sleeve situated on the rod that axially moves relative to the rod in response to the reception into and retraction from the handle assembly head, with the sleeve controlling capture and release of the implant by the rod. The distal end of the rod has a clamp formed by jaws that flex to capture and release the implant.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This U.S. non-provisional utility patent application claims the benefit of and/or priority under 35 U.S.C. § 119(e) to U.S. provisional patent application Ser. No. 63/300,925 filed Jan. 19, 2022 titled “Two-Piece Spine Implant Installation Instrument For Use With An Endoscope And Method of Use,” the entire contents of which is specifically incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to methods and instruments for endoscopic spine surgery and, more particularly, to methods and instruments for endoscopically installing spine implants, such as but not limited to interbody spine implants, during a surgical spine procedure.

BACKGROUND OF THE INVENTION

(2) Many people contend with spine issues due to age, disease, trauma, congenital, and acquired complications and conditions. While some spine issues can be alleviated without surgery, other spine issues necessitate surgery. Spine surgery may entail removing bone, tissue, and/or other material from a spine; replacing a portion or portions of a spine with one or more spine implants;

affixing various spine hardware to the spine typically for stabilizing or providing fixation of one or more portions or elements of a spine; and/or installing a spine implant into and/or onto a spine.

(3) It is important to minimize the amount of tissue injury, trauma, and post-operative pain to optimize a patient's recovery from spine surgery. Advances in spine surgery technology, equipment, instrumentation and procedures (collectively, spine surgery) over the years have resulted in significant improvement in lessening tissue injury, trauma, and post-operative pain. One advance was Minimally Invasive Surgery (MIS), where spinal surgical instruments and/or spine implants and other hardware are introduced into the body via cannulas that extend through a small incision in the body and positioned accordingly. Multiple cannulas with various surgical spine instruments are generally used. A currently foremost advance is Endoscopic Spine Surgery (ESS) that uses micro-sized incisions (less than 1-inch) and small tubular systems in combination with an endoscope to visualize the surgical field.

(4) Particularly, ESS employs an endoscope to introduce a spine instrument and/or spine implant into the body and to perform a spine procedure using same. An endoscope is a slender, tubular optical instrument used as a viewing system for examining a body part with an associated spine instrument. With advances in optics, visualization of tissues, spinal imaging, and overall greater precision, endoscopic surgery is well suited for various spine procedures. One type of ESS may entail installing an implant between adjacent vertebrae of a spine (i.e., an intervertebral or interbody spine implant) as, for example, a substitute for a removed vertebral disc. The interbody spine implant installation instrument must provide for accurate placement, orientation, and possible deployment of an interbody spine implant within the constraints of an endoscope.

(5) Current installation instruments for installing spine implants, such as but not limited to interbody spine implants, via an endoscope are wont in many respects. It would therefore be advantageous to have an instrument for installing a spine implants such as but not limited to interbody spine implants via an endoscope that addresses problems of prior art endoscopic spine implant installation instruments. It would furthermore be advantageous to have a method for installing spine implants such as but not limited to interbody spine implants via an endoscope.

(6) The present two-piece spine implant installation instrument for use with an endoscope and methods of use addresses the above and more.

SUMMARY OF THE INVENTION

(7) A medical instrument for installing a spine implant, such as, but not limited to, an expandable interbody or intervertebral spine implant, via an endoscope during a surgical spine procedure, utilizes a two-piece construction to deliver and insert the spine implant. The medical instrument or spine implant installer or installation tool is characterized by a first piece (handle assembly) and a second piece (shaft assembly) connectable to the handle assembly and sized for extension through the endoscope. The shaft assembly has a clamp at a distal end thereof that is configured to hold a preferably, but not necessarily, expandable interbody spine implant when a proximal end of the shaft assembly is fully received by a head of the handle assembly, and to release the expandable interbody spine implant upon retraction of the shaft assembly from the head of the handle assembly. Reception and retraction of the shaft assembly by the handle assembly is accomplished by manipulation of a rotatable knob of the head.

(8) The shaft assembly has a rod with a longitudinal bore and a sleeve situated on the rod that axially moves relative to the rod in response to the reception of the rod (shaft assembly) into and retraction from the head of the handle assembly with the sleeve controlling capture and release of an expandable interbody implant by the clamp of the rod. The rod and sleeve share the same outer diameter.

(9) The rod defines a proximal rod end and a distal rod end, with the proximal rod end having external threading and the distal rod end having the clamp formed by jaws that flex to capture and release the interbody spine implant. The sleeve has a distal sleeve end with a boss sized to receive the jaws of the distal end of the rod during reception of the rod into the handle assembly, and tangs

that extend from the boss along the outside of the rod to near a proximal end of the sleeve. A spring about the jaws of the rod adjacent the boss aids in releasing the interbody spine implant from the clamp of the rod when the clamp emerges from the boss of the sleeve through axial movement of the rod.

(10) The handle assembly has a handle portion defining a proximal handle portion end and a distal handle portion end with the head of the handle assembly connected to the proximal end, the head defined by a frame that includes an orifice and the rotatable knob which receives the proximal rod end. The proximal rod end of the rod of the shaft assembly is received into and removed from a nozzle of the frame that directs the proximal rod end into the rotatable knob.

(11) In use, the portion of the installer that clasps onto the expandable interbody spine implant is fed up through the bottom of the cannulated endoscopic tube (endoscope) and engages with the handle assembly once fully inserted. As the threaded knob of the handle assembly is rotated, it draws the rod of the shaft assembly with its associated clasping jaws back into the handle assembly, while at the same time the tangs of the sleeve of the shaft assembly (on the rod of the shaft assembly) recede into the handle assembly and bottom out on flats within the handle assembly. This in turn causes the sleeve to compress the clasping jaws and lock onto the expandable interbody spine implant, all while being inserted through the endoscope. Once the expandable interbody spine implant has been inserted, the knob is rotated to back out the rod from the handle assembly, causing the spring to expand the jaws once the jaws emerge from the boss to release the expandable interbody spine implant.

(12) Further aspects of the present invention will become apparent from consideration of the drawings and the following description of forms of the invention. A person skilled in the art will realize that other forms of the invention are possible and that the details of the invention can be modified in a number of respects without departing from the inventive concept. The following drawings and description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention and its features, functions, and components will be better understood by reference to the accompanying drawings, wherein:

(2) FIG. 1 is a front perspective view of an exemplary two-piece spine implant installation instrument (medical instrument), shown disassembled into a handle assembly and a shaft assembly, along with an endoscope, the shaft assembly holding an exemplary expandable interbody (intervertebral) spine implant;

(3) FIG. 2 is a front perspective view of the shaft assembly of the present two-piece spine implant installation instrument;

(4) FIG. 3 is an enlarged side sectional view of a distal end of the shaft assembly of the present two-piece spine implant installation instrument;

(5) FIG. 4 is an enlarged rear view of the distal end of the shaft assembly of the present two-piece spine implant installation instrument;

(6) FIG. 5 is a rear perspective view of the shaft assembly of the present two-piece spine implant installation instrument;

(7) FIG. 6 is a front perspective view of the shaft assembly of the present two-piece spine implant installation instrument along with the associated endoscope;

(8) FIG. 7 is an enlarged front perspective view of the shaft assembly of the present two-piece spine implant installation instrument situated in and through the associated endoscope;

(9) FIG. 8 is a front perspective view of the handle assembly of the present two-piece spine implant installation instrument;

(10) FIG. 9 is an enlarged front perspective view of the handle assembly of the present two-piece spine implant installation instrument;

(11) FIG. 10 is a front perspective view of the present two-piece spine implant instrument, shown assembled, along with the associated endoscope;

(12) FIG. 11 is an enlarged perspective view of a distal end of the handle assembly of the present two-piece spine implant installation instrument and a proximal end of the shaft assembly of the present two-piece spine implant installation instrument extending from the associated endoscope and about to be received in the distal end of the handle assembly;

(13) FIG. 12 is a sectional view along a longitudinal axis of the shaft assembly of the distal end of a head of the handle assembly of the present two-piece spine implant installation instrument and a proximal end of the shaft assembly of the present two-piece spine implant installation instrument extending from the associated endoscope and about to be received in the distal end of the head of the handle assembly;

(14) FIG. 13 is the sectional view of FIG. 12 showing the proximal end of the shaft assembly of the present two-piece spine implant installation instrument received into the distal end of the head of the handle assembly;

(15) FIG. 14 is an enlarged view of the sectional view of FIG. 12 showing the proximal end of the shaft assembly of the present two-piece spine implant installation instrument received into the distal end of the head of the handle assembly;

(16) FIG. 15 is an enlarged perspective of the exemplary expandable interbody spine implant received on the distal end of the head of the handle assembly and extending from the associated endoscope;

(17) FIG. 16 is a reduced perspective view of the exemplary expandable interbody spine implant as received on the distal end of the head of the handle assembly and extending from the associated endoscope as depicted in FIG. 15 showing a longitudinal axis of the shaft assembly;

(18) FIG. 17 is a perspective view of the distal end of the head of the handle assembly attached to the shaft assembly that extends from the associated endoscope;

(19) FIG. 18 is a sectional view along a longitudinal axis of the distal end of the head of the handle assembly, shaft assembly, and endoscope of FIG. 17 with the proximal end of the shaft assembly received by the head of the handle assembly, and a proximal end of a first shaft of the shaft assembly received in a control knob of the head of the handle assembly;

(20) FIG. 19 is a reduced perspective view of the present two-piece spine implant installation instrument holding the exemplary expandable interbody spine implant and associated endoscope;

(21) FIG. 20 is the reduced perspective view of the present two-piece spine implant installation instrument of FIG. 19 with the endoscope and shaft assembly shown exploded from the handle assembly and the exemplary expandable interbody spine implant;

(22) FIG. 21 is a reduced perspective view of the present two-piece spine implant installation instrument from a side opposite that shown in FIG. 20;

(23) FIG. 22 is a perspective view of the distal end of the present two-piece spine implant installation instrument with the exemplary expandable interbody spine implant received by a clamp of the distal end of the shaft assembly; and

(24) FIG. 23 is a front perspective view of the present two-piece spine installation instrument shown without the exemplary expandable interbody spine implant.

(25) For the purposes of promoting an understanding of the principles of the invention reference will now be made to the embodiment illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the invention is thereby intended. Any alterations and further modifications in the described embodiment, and any further applications of the principles of the invention as described herein are contemplated as would normally occur to one skilled in the art to which the invention relates.

DETAILED DESCRIPTION OF THE INVENTION

(26) FIGS. 1-23 depict various views of the present two-piece interbody or intervertebral spine implant installation instrument (medical instrument), generally designated **10**, and/or its constituent elements and components—including a handle assembly **12** and a shaft assembly **14**, along with an exemplary endoscope **200** for which the two-piece intervertebral spine implant installation instrument **10** is designed to use. The endoscope **200** is representative of all types of endoscopes with which the two-piece interbody spine implant installation instrument **10** may be used. As is typical, the endoscope **200** has a tubular body **202** with a longitudinal bore **204** defining a distal endoscope end **205** and a proximal endoscope end **207**, along with optics **206**. The endoscope **200** may have other features and/or shapes not shown. The two-piece interbody spine implant installation instrument **10** is made from one or more surgical grade materials.

(27) FIGS. 1 and 10 depict the two-piece interbody spine implant installation instrument **10**, with FIG. 1 showing the two-piece interbody spine implant installation instrument **10** in an exploded view, and FIG. 10 showing the two-piece interbody spine implant installation instrument **10** assembled, both holding an exemplary expandable interbody spine implant **250** along with an endoscope **200**. The exemplary expandable interbody spine implant **250** is representative of all types of interbody spine implants (expandable and non-expandable) as well as other spine implants. FIG. 2 depicts an enlarged view of the shaft assembly **14** of the two-piece interbody spine implant installation instrument **10**. The shaft assembly **14** is characterized by a rod or the like **40** having a longitudinal bore **42**, an externally threaded proximal end **41**, and a distal end **46** with a clamp/clamp structure **48**. A first flat **43** extends along the outside longitudinal length of the rod **40** (of which only a portion of the first flat **43** can be seen) that is proximate the externally threaded proximal end **41**, while a second flat **44** (see, e.g., FIG. 5) extends along the outside longitudinal length of the rod **40** opposite the first flat **43**, the nomenclature first and second being arbitrary. The first and second flats **43**, **44** extend through the external threading of the proximal end **41**. The distal clamp **48** of the rod **40** is defined, at least in part, by a first flange **51** having a first lip **52** that projects radially inward at the proximal end of the first flange **51**/clamp **48**, and a second flange **53** having a second lip **54** that projects radially inward at the proximal end of the second flange **53**/clamp **48**, the nomenclature first and second being arbitrary.

(28) A sleeve **55** is disposed on and/or about the rod **40** for axial movement thereof relative to the rod **40**. As best discerned in FIGS. 3-5, the distal end of the sleeve **55** has a boss **47** defined by a first collar **49** and a second collar **50** having a diameter larger than the first collar **49**, the nomenclature first and second being arbitrary. The second collar **50** has a cavity or socket **65** that is sized to receive at least a rear portion or more of the clamp **48** of the rod **40** and, more particularly, the stepped rear (distal) ends of the first and second flanges **51**, **53** of the clamp **48**. The sleeve **55** also has a first tang **56** that extends along the first flat **43** of the rod **40** from the distal clamp **48** to a proximal end **58** thereof, and a second tang **60** that extends along the second flat **44** of the rod **40** from the distal clamp **48** to a proximal end **62** thereof. The first tang **56** has a bore **57** situated proximate the proximal end **58** of the first tang **56**. The second tang **60** also has a bore **61** situated proximate the proximal end of the second tang **60**. The rod **40** and sleeve **55** share the same outer diameter. This allows the shaft assembly **14** to fit through the endoscope and not be restricted in axial movement thereof on the rod **40**. A spring or spring structure (spring) **64** is situated on and/or about a neck of the first and second flanges **51**, **53**. The spring **64** is formed to outwardly bias (spread out) the first and second flanges **51**, **53** such that the clamp **48** is normally biased to release any implant and be ready to grasp an implant.

(29) FIG. 6 depicts the shaft assembly **14** that is not holding an interbody spine implant ready to be inserted into the endoscope **200**. FIG. 7 is an enlarged view of the shaft assembly **14** and endoscope **200** of FIG. 6 but with the shaft assembly **14** fully inserted into and through the endoscope **200**. The assemblage (FIG. 7) is ready for reception by the handle assembly **12**.

(30) Referring to FIGS. 8 and 9, the handle assembly **12** is shown. The handle assembly **12** has a handle **16** and a head **20**. The handle **16** has a bottom (distal end) **17** and a top (proximal end **18**),

the handle **16** shaped to ergonomically fit a hand. A stem **19** extends from the top **18** that supports the head **20**. The head **20** has a distal end **21** with a configured notch **29**, and a proximal end **22**. A neck **23** extends from the distal end **21** and supports a frame **24** having an upper frame portion **24a** and a lower frame portion **24b**, a nose **25**, and a stop **28** opposite the nose **25**. The stop **28** is part of first and second lateral rails **31**, **32** that extend from and between the distal end of the upper and lower frame portions **24a**, **24b** that may be translatable on the upper and lower frame portions **24a**, **24b** in a direction co-axial with a drive axis DA of the head **20**. The stop includes a ring of teeth/gearing **30**. The head **20** is generally transverse to the handle **16** (see, e.g., FIGS. **1** and **8**). (31) A knob **27** is situated in the frame **24** between the nose **25** and the stop **28**. The knob **27** is rotatable within the upper frame portion **24a** and the lower frame portion **24b** (see, e.g., FIG. **18**) and has an internally threaded bore **38** (see, e.g., FIGS. **12-14**) that extends along the axis of rotation of the knob **27** (which is co-axial with the drive axis DA of the head **20**, which is co-axial with the longitudinal axis LA of the shaft assembly **14**) and through the knob **27**. The internally threaded bore **38** is threaded and sized to receive the externally threaded proximal end **41** of the rod **40**. The proximal end **22** of the head **20** has a generally round nozzle **26** (although other shapes may be used along with the shape of the rod **40**) that provides an axial opening into the frame **24** and into the internally threaded bore **38** of the knob **27** so the knob **27** can receive the rod **40**. A first flat **34** is provided in the outside of the nozzle **26** (see, e.g. FIGS. **11-14**), while a second flat **35** is provided in the outside of the nozzle **26** (see, e.g. FIGS. **11-14**), the nomenclature first and second being arbitrary. Rotation of the knob **27** in one direction draws the proximal end **41** of the rod **40** therein while rotation of the knob **27** in another opposite direction releases the proximal end **41** of the rod **40** from the knob **27**.

(32) Reception of the shaft assembly **14** into the head **20** causes the clamp **48** of the shaft assembly **14** to close its jaws and grasp, hold or the like an interbody spine implant such as the exemplary expandable interbody spine implant **250** (see, e.g., FIGS. **15-16**) by axially pulling the clamp **48** into the socket **65** of the boss **47** through mechanical restriction of axial translation of the sleeve on and relative to the rod **40** as the rod **40** axially translates. In FIG. **11**, the threaded proximal end **41** of the rod **40** of the shaft assembly **14** is shown extending from the proximal end **207** of the endoscope **200** ready for reception into the distal end **22** (nozzle **26**) of the head **20** in order for the clamp **48** to hold an implant. In FIG. **17**, the threaded proximal end **41** of the rod **40** of the shaft assembly **14** has been initially received into the distal end **22** (nozzle **26**) of the head **20**. FIGS. **12-14** and **18** additionally depict the sequence of reception of the threaded proximal end **41** of the rod **40** from the endoscope body **202** and into the threaded bore **38** of the knob **27**. In FIG. **12** the threaded proximal end **41** of the rod **40** is adjacent the nozzle bore **26** with the first tang **56** in the slot **43** of the rod **40** situated distal to the end **41**, and the second tang **60** in the slot **44** of the rod **40** situated distal to the end **41**. This is also seen in FIG. **23**. In FIG. **13**, the end **41** has been received into the nozzle **26** and is being introduced into the threaded bore **38** of the knob **27**. Rotation of the knob **27** draws the rod end **41** axially into the threaded bore **38**. As discerned in FIG. **13**, the tangs **56** and **60** of the sleeve **55** are now adjacent the distal end **22** of the head **20**. In FIGS. **14** and **18**, the rod end **41** has been drawn further into the threaded bore **38** such that the end **58** of the first tang **56** of the sleeve **55** and the end **62** of the second tang **60** of the sleeve **55** both contact an annular ledge of the nozzle (see FIG. **18**) which prevents further axial movement of the sleeve **55** on the rod **40** as the rod **40** is further axially moved into threaded bore **38** during further rotation of the knob **27** (in the same rotational direction).

(33) The clamp **48** is configured to receive and hold an interbody spine implant, such as the exemplary expandable interbody spine implant **250**, as the clamp **48** is axially drawn into the boss socket **65** upon longitudinally axial movement of the rod **40** (along the longitudinal axis LA of the rod **40**—see, e.g., FIGS. **16** and **18**) relative to the sleeve **55**, due to the sleeve being stopped from further axial movement by the ends **58**, **62** of the sleeve tangs **56**, **60** abutting the internal annular ring of the nozzle **26** (as sequenced in FIGS. **11-18**). The rod **40**, however, can still advance in the

threaded knob bore **38** (see, e.g., FIG. **17**). As the rod **40** advances further into the knob bore **38** through further knob rotation, the first and second flanges **51**, **53** are pulled axially into the boss socket **65** where, against the bias of the spring **64**, the first and second flanges **51**, **53** are forced to flex radially inward, thereby grasping an implant (see, e.g., FIGS. **15**, **16**). Both FIGS. **15-17** depict a position wherein the rod **40** has been fully drawn into the knob **27**, the clamp **48** is securely holding the implant **250**, and the clamp **48** has been fully pulled into the socket **65** of the boss **50**.

(34) Referring additionally to FIGS. **19-22**, a method of use of the present two-piece interbody spine implant installation instrument **10** in conjunction with an endoscope can be discerned and herein described. Other methods and/or sequences are contemplated. In use, the portion of the installer that clasps onto the expandable interbody spine implant (e.g., clamp **48**) is fed up through the bottom of the cannulated endoscopic tube (e.g., endoscope **200**) and engages with the handle assembly (e.g., handle assembly **12**) once fully inserted. As the threaded knob (e.g., knob **27**) of the handle assembly is rotated, it draws the rod (e.g., rod **40**) of the shaft assembly (e.g., shaft assembly **14**) with its associated clasping jaws (e.g., jaws **51**, **53**) back into the handle assembly, while at the same time the tangs (e.g., tangs **56**, **60**) of the sleeve (e.g., sleeve **55**) of the shaft assembly (on the rod of the shaft assembly) recede into the handle assembly and bottom out on flats within the handle assembly. This in turn causes the sleeve to compress the clasping jaws and lock onto the expandable interbody spine implant (e.g., implant **250**), all while being inserted through the endoscope. Once the expandable interbody spine implant has been inserted, the knob is rotated to back out the rod from the handle assembly, causing the spring to expand the jaws once the jaws emerge from the boss to release the expandable interbody spine implant.

(35) While the invention has been illustrated and described in detail in the drawings and foregoing description, the same is to be considered as illustrative and not restrictive in character, it being understood that only a preferred embodiment has been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected. It should be understood that while the use of words such as preferable, preferably, preferred or more preferred utilized in the description above indicate that the feature so described may be more desirable, it nonetheless may not be necessary and embodiments lacking the same may be contemplated as within the scope of the invention.

Claims

1. A medical instrument for placing an interbody spine implant in an intervertebral space of a spine with the use of an endoscope, the medical instrument comprising: a first part including a handle, a head on the handle, an orifice in the head, and a rotatable knob situated in the head, the rotatable knob having an internally threaded bore axially aligned with the orifice; a second part operably connectable to the head on the handle and configured for receipt through an endoscope, the second part including a rod with a longitudinal bore and defining a distal rod end and a proximal rod end with the proximal rod end having external threading, a clamp at the distal rod end and configured to hold an intervertebral spine implant when the externally threaded proximal rod end is fully threaded into the internally threaded bore of the rotatable knob of the head and through the orifice of the head on the handle, and to release the intervertebral spine implant upon unthreading of the externally threaded proximal rod end from the internally threaded bore of the rotatable knob of the head on the handle, and a sleeve situated on the rod that axially moves relative to the rod in response to threaded reception into and unthreaded retraction from the rotatable knob of the head of the handle, wherein the clamp is formed by jaws that flex to capture and release the intervertebral spine implant, wherein the clamp is formed by jaws that flex to capture and release the intervertebral spine implant; the sleeve defining a distal sleeve end with a boss sized to receive the jaws of the distal rod end during threaded reception of the externally threaded proximal end of the rod into the orifice of the head, and a proximal sleeve end, and tangs that extend from the boss

along an outside of the rod to near the proximal sleeve end; and a spring about the jaws of the rod adjacent the boss to aid in releasing the intervertebral spine implant from the clamp of the rod when the clamp emerges from the boss of the sleeve through axial movement of the rod.

2. The medical instrument of claim 1, wherein the handle defines a proximal handle end and a distal handle end with the head of the handle connected to the proximal handle end, the head defined by a frame that includes the orifice and the rotatable knob.

3. The medical instrument of claim 2, wherein the jaws comprise a first jaw portion and a second jaw portion.

4. The medical instrument of claim 3, wherein the handle is disposed perpendicular to the head.

5. The medical instrument of claim 4, wherein the head is connected to the proximal handle end via a neck.
